

Global CO₂ Emissions Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview of how fossil fuel + industry CO₂ evolved from the Industrial Revolution to 2024, highlighting regional share shifts, per capita disparities, and the latest annual-change signals.

35+ Bt

Global annual CO₂ from fossil fuels + industry today (exceeds 35 billion tonnes/year).

150x

Per-capita gap between highest and lowest emitters (e.g., US/Australia vs. Chad/Niger).

The February 2026 Mauna Loa reading is more than **50% above** the pre-industrial baseline (~280 ppm).

Increase relative to pre-industrial (~280 ppm → 429 ppm).

1958 → today: sharp acceleration (measurement era)

1958 2026

Visual cue only (not a quantified timeseries). Uses the “since 1958” acceleration framing in the brief.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold

From ~6 billion tonnes in 1950 to 35+ billion tonnes today — with major inflection points in the late 20th century.

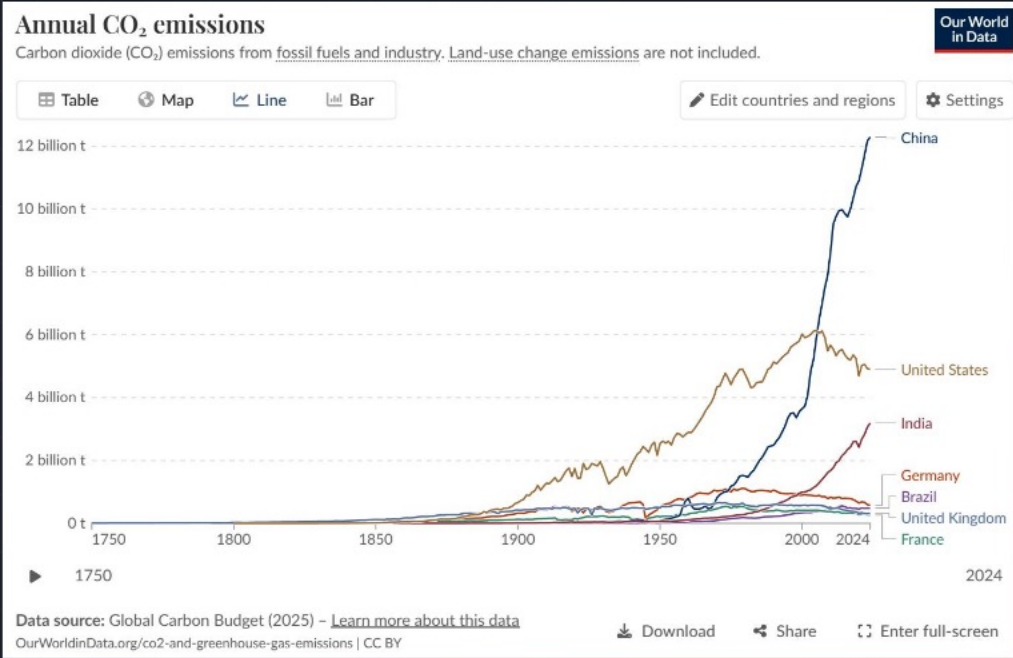
6 → 20 → 35+

1950 (~6 Bt) · 1990 (>20 Bt) · 2024 (>35 Bt).

- Growth accelerated after mid-century industrial expansion.
- Recent growth has slowed — but emissions have not yet peaked.

Figure reference: figure_2.jpg (line chart of global emissions, 1750–2024).

FIGURE: 1750–2024 LINE CHART



Rendered as provided (placeholder if source image differs).

China Now Emits ~12.5 Bt — More Than Double the US

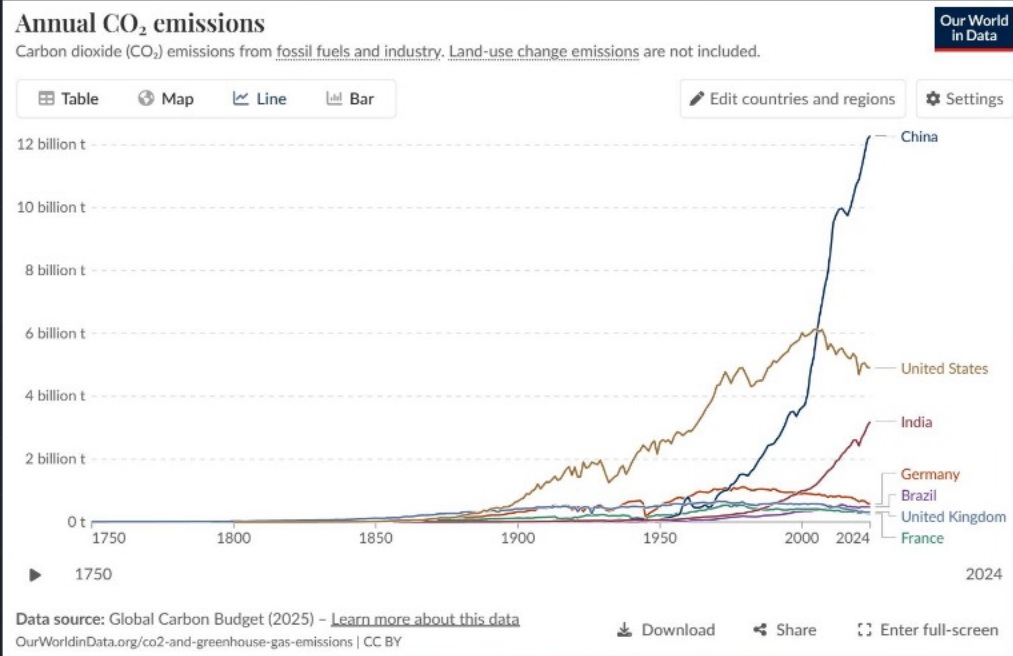
Annual fossil fuel + industry CO₂: China (~27% of global) vs. United States (~14%). India is third and rising.

~27% vs ~14%

Share of global emissions: China (~27%) vs. United States (~14%).

- China’s rise steepened after 2000.
- US peaked near ~6 Bt around 2005; now ~5 Bt and declining.
- EU-27 ~2.7 Bt (~7%) and declining; India ~3 Bt (~8%) rising.

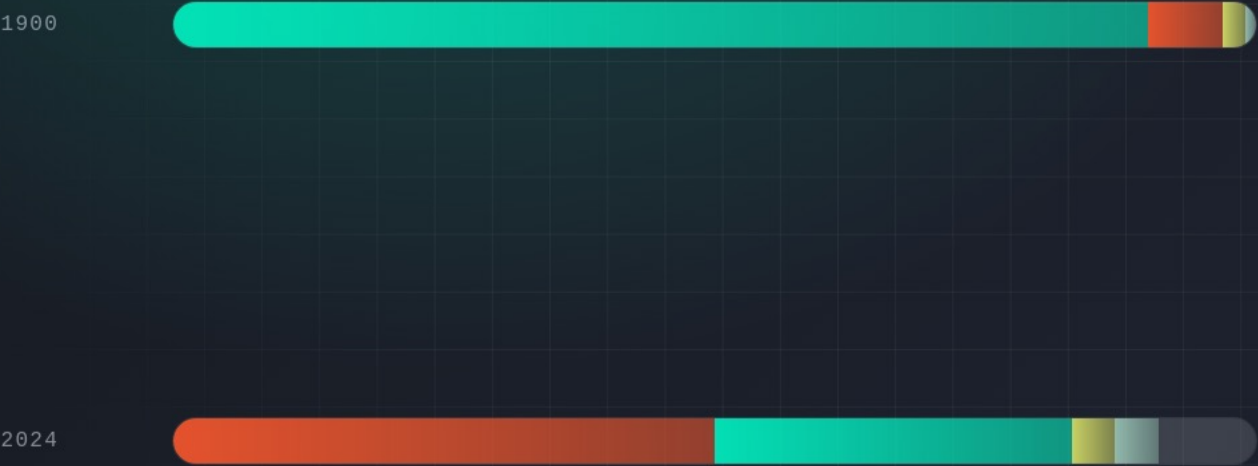
FIGURE: ANNUAL CO₂ BY COUNTRY



Contains the country-level line chart referenced in the brief.

Asia Now Produces ~50% of Global Emissions

The center of emissions shifted from Europe + the US dominating in 1900 to Asia representing roughly half today.



■ US + Europe ■ Asia ■ Africa ■ South America

Visual is a schematic side-by-side share comparison (not a full stacked area chart). Values reflect the brief: Europe+US >90% (1900) → US+Europe < one-third (2024), Asia ~50%, Africa/South America 3-4% each.

NEGOTIATION RELEVANCE

< 33%

US + Europe combined share today (below one-third), despite large historical contributions.

- Africa and South America remain low contributors (3-4% each).
- These shares are comparable to international aviation + shipping combined (as noted in the brief).

Use this slide to frame “current vs. historical responsibility” discussions.

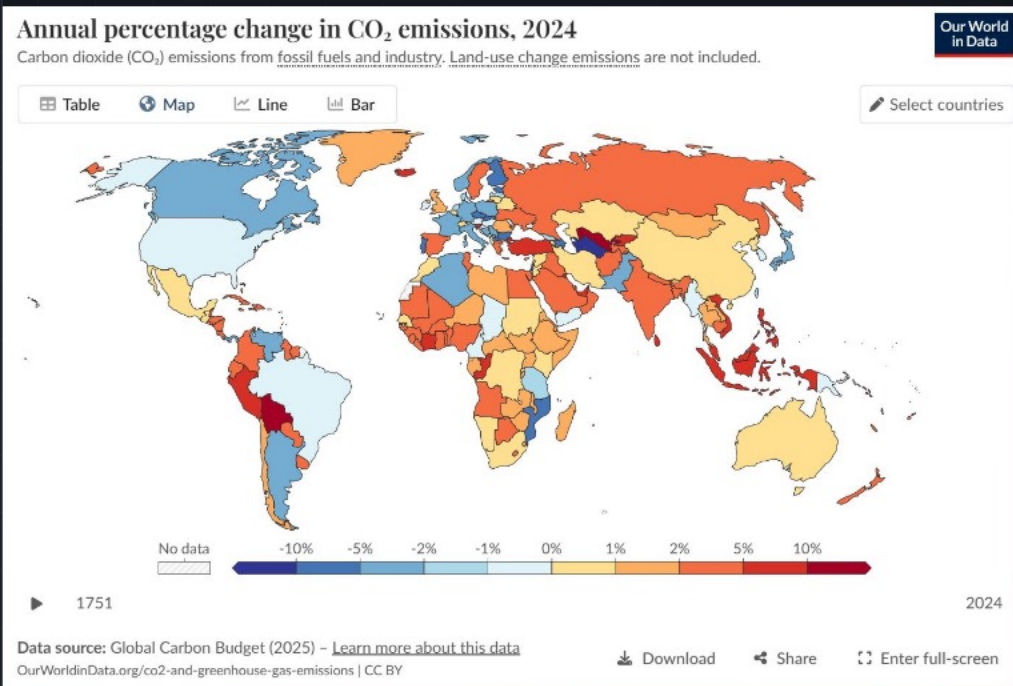
Per Capita Gap: ~15 t (US) vs ~0.1 t (Chad/Niger) — 150x

Horizontal bars show selected countries (tonnes CO₂ per person per year). Max scale anchored at 15 t.



Note: The brief states the “150x” gap and the “under two days vs. a year” framing for US/Australia vs. Chad/Niger.

FIGURE: PER CAPITA CO₂ OVER TIME



Use alongside bars to connect “levels” with historical trajectories.

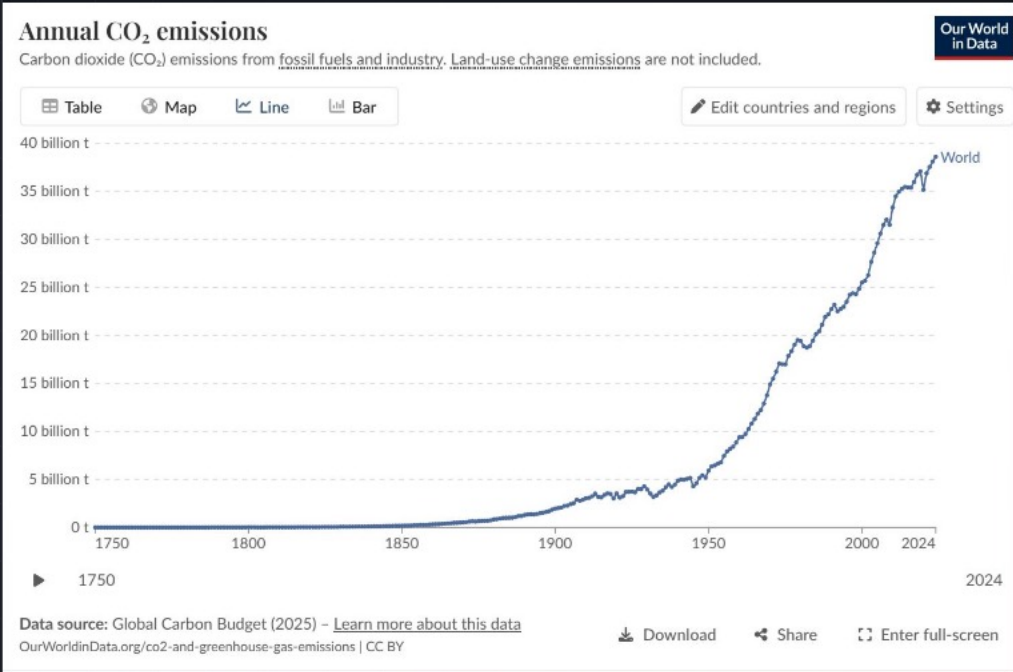
2024 Snapshot: US + Europe Down; Russia + SE Asia Up

Annual percentage change map shows a mixed pattern: blue (declines) across the US and much of Europe, orange/red (increases) across Russia, parts of Southeast Asia, and several African nations.

- US and much of Europe: roughly -1% to -5% (blue).
- Russia, parts of SE Asia, several African nations: +2% to +10% or more (orange/red).
- South America: one or two countries show very large increases (>10%), while others decline.

Interpretation focuses on sign + broad ranges as stated in the brief.

FIGURE: % CHANGE MAP (2024)



Map legend is in the source image.

Energy Mix Explains Diverging Per-Capita Outcomes

The brief notes that France and the UK emit far less per capita than Germany or the Netherlands, largely due to higher nuclear + renewable shares versus higher fossil dependence.

Policy + power

Per-capita CO₂ is not determined by prosperity alone.

- France and UK: lower per-capita CO₂ (~4 t in the brief table) with cleaner electricity mix.
- Germany/Netherlands: higher fossil share (qualitative comparison in brief).

COMPARISON TABLE (QUALITATIVE)

COUNTRY GROUP	ELECTRICITY MIX (AS DESCRIBED)	PER-CAPITA CO ₂ (FROM BRIEF)
France / UK	Higher nuclear + renewables (lower fossil share)	~4 t/person/year
Germany / Netherlands	Higher fossil fuel share (relative to France/UK)	Higher than France/UK (qualitative; not quantified in brief)

The brief provides a qualitative energy-mix explanation but does not provide numeric energy shares or Netherlands per-capita values.

Key Takeaways: Emissions Still Rising; Responsibility Map Shifting

- Global fossil fuel + industry CO₂ exceeds **35 Bt/year** and has not yet peaked.
- China accounts for **over 25%** of global emissions (~12.5 Bt), more than double the US (~5 Bt).
- Per-capita inequalities remain extreme: roughly **150×** between high and low emitters.
- Geographic center shifted: from >90% Europe+US (1900) to Asia at ~50% today; US+Europe below one-third.
- Energy mix and policy choices matter: similar prosperity can yield very different per-capita CO₂ outcomes.

All bullets sourced from the “Key Findings” section in the brief.

● Use totals + shares for mitigation allocation

● Use per-capita to frame equity implications

● Use 2024 change to discuss near-term trajectory